

Information theory: surprise, entropy and codes

An outcome with probability p carries $-\log_2 p$ bits. Entropy averages that surprise, and no code can use fewer bits per symbol on average.

THE QUESTION

How many bits does data from a source p need, and what does a wrong model q of it cost?

THE INTUITION

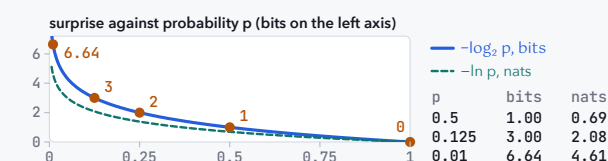
Rare outcomes tell us more, so surprise is $-\log p$. Its average, the entropy, is the floor for the average code length. Lengths built for q cost the cross-entropy; the excess is $\text{KL}(p \parallel q)$.

KEY TAKEAWAY

$H(p) \leq L < H(p) + 1$ for the best prefix code. $H(p, q) = H(p) + \text{KL}(p \parallel q)$ with $\text{KL} \geq 0$. Minimising cross-entropy loss is maximum likelihood.

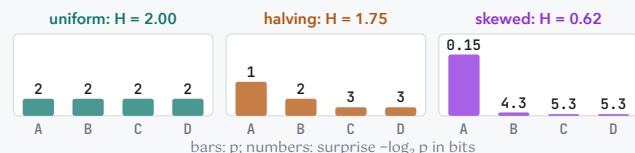
surprise $-\log p(x)$ → $H(p)$: average surprise → code built for q : $H(p, q)$ → $\text{KL} = H(p, q) - H(p)$ → $I(X; Y) = \text{KL}(\text{joint} \parallel \text{product})$

1 Surprise is $-\log p$, in bits or nats



A surprise score should fall as $p(x)$ rises, be 0 at $p(x) = 1$ and add for independent events. As $p(x, y) = p(x)p(y)$, a logarithm is the one choice: $I(x) = -\log p(x)$, so $I(x, y) = I(x) + I(y)$. Base 2 gives bits, base e nats; 1 nat = 1.443 bits. An outcome with $p = 1/8$ carries 3 bits, the answers to three fair yes/no questions. Snow in a warm city is rarer than snow in the mountains, so it surprises more.

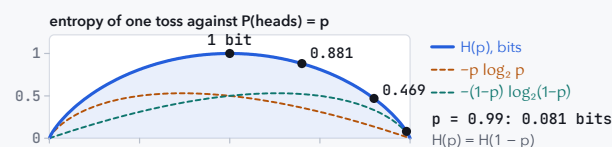
2 Entropy is the average surprise



$$H(X) = \mathbb{E}_{x \sim p}[-\log_2 p(x)] = -\sum_{i=1}^k p_i \log_2 p_i, \quad 0 \log 0 = 0$$

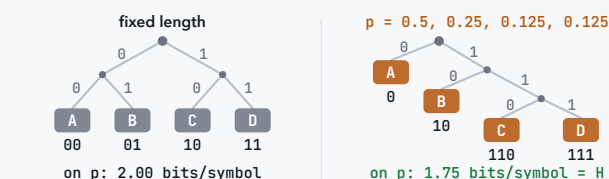
It runs from 0 (one sure outcome) to $\log_2 k$ (uniform): 2 bits for four symbols, or $\ln 4 = 1.386$ nats. The slides' three sources give exactly 2, 7/4 and 0.619 bits. Only the probabilities count, not the labels or values.

3 One coin toss carries at most one bit



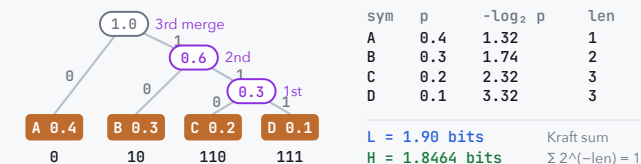
$H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ is concave, with its maximum of 1 bit at $p = 0.5$. A coin with $P(\text{heads}) = 0.9$ gives 0.469 bits per toss, so long runs of it compress to under half a bit per toss. In nats the maximum is $\ln 2 = 0.693$. The same two terms make up the binary cross-entropy loss (panel 10).

4 Prefix codes give likely symbols short words



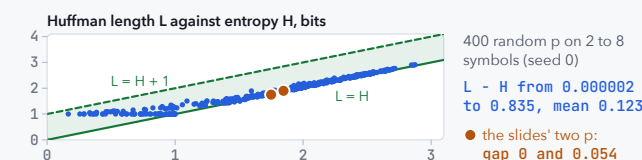
Average length: $L = \sum_i p_i l_i$. The fixed code spends 2 bits on every symbol; the code 0, 10, 110, 111 spends 1.75, equal to H , because each $l_i = -\log_2 p_i$ exactly. Its words are the leaves of a binary tree, so none starts another (a prefix code) and a stream needs no separators: 0101100111 reads 0|10|110|111 = ABCAD.

5 Huffman: merge the two least likely, repeat



Each merge adds one bit to every word below it. For the slides' $p = (0.4, 0.3, 0.2, 0.1)$ the lengths are 1, 2, 3, 3 and $L = 1.9 > H = 1.8464$: the ideal lengths $-\log_2 p_i$ are not whole numbers. Huffman has the smallest L of any prefix code, and meets H only when every p_i is a power of $1/2$. Prefix codes obey Kraft's inequality, $\sum_i 2^{-l_i} \leq 1$. The notebook's builder, run on three equally likely symbols, would give lengths 1, 2, 2: $L = 5/3$ against $\log_2 3 = 1.585$.

6 No code beats H ; Huffman is within a bit

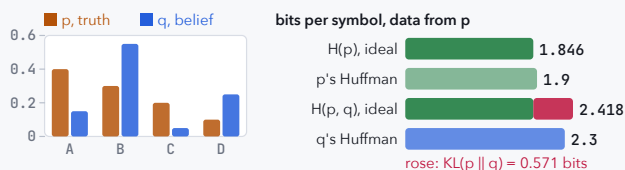


$H(p) \leq L_{\text{best}} < H(p) + 1$. Lower bound: with $r_i = 2^{-l_i}$, Kraft and Gibbs' inequality (panel 8) give $L - H = \sum_i p_i \log_2 (p_i / r_i) \geq 0$. Upper bound: the lengths $\lceil -\log_2 p_i \rceil$ obey Kraft and cost less than $H + 1$. Coding blocks of n symbols gets within $1/n$ bit of H per symbol.

A wrong model costs KL extra bits

Cross-entropy is what q 's code costs on data from p ; KL is the excess over $H(p)$. Mutual information is the KL between a joint and the product of its marginals.

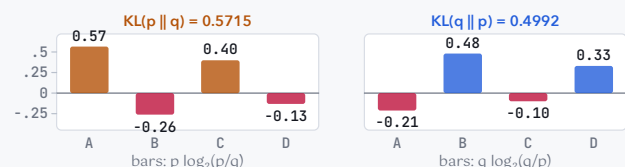
7 Cross-entropy: data from p , lengths from q



$$H(p, q) = \mathbb{E}_{x \sim p}[-\log_2 q(x)] = -\sum_i p_i \log_2 q_i \geq H(p)$$

The slides' halving source sent with a uniform belief's 2-bit code costs 2 bits, not 1.75. Their second pair: $H(p) = 1.8464$, $H(p, q) = 2.4179$ (notebook agrees). Real codes round lengths: q 's Huffman code (A 3, B 1, C 3, D 2 bits) costs 2.3 on p , below $H(p, q)$. Only $H(p)$ bounds every code.

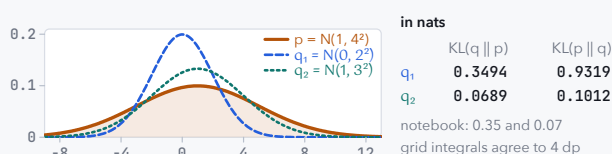
8 KL is never negative and not symmetric



$$KL(p \parallel q) = \sum_i p_i \log_2 \frac{p_i}{q_i} = H(p, q) - H(p)$$

By Jensen, $-KL = \sum_i p_i \log(q_i/p_i) \leq \log \sum_i q_i = 0$, with equality only at $q = p$ (Gibbs' inequality). Single terms go negative; the sum cannot. The two directions differ, 0.5715 against 0.4992 bits, and $q_i = 0$ where $p_i > 0$ makes $KL(p \parallel q)$ infinite. It is not a distance.

9 Gaussians have a closed-form KL



For $q = \mathcal{N}(\mu_1, \sigma_1^2)$ and $p = \mathcal{N}(\mu_2, \sigma_2^2)$, in nats,

$$KL(q \parallel p) = \ln \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

So $KL(q_1 \parallel p) = \ln 2 + 5/32 - 1/2 = 0.3494$, larger than $KL(q_2 \parallel p)$, as the notebook finds (0.35, 0.07). Reversed, q_1 costs 2.7 times more. Which direction to fit when q approximates a posterior (mode seeking or mass covering) is on the variational inference sheet.

10 Cross-entropy loss is maximum likelihood

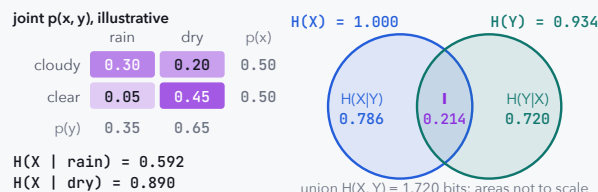


For data $x_1, \dots, x_N$ with empirical distribution $\hat{p}$, the mean negative log-likelihood is a cross-entropy:

$$-\frac{1}{N} \sum_n \log q_\theta(x_n) = H(\hat{p}, q_\theta) = H(\hat{p}) + KL(\hat{p} \parallel q_\theta)$$

$H(\hat{p})$ is fixed, so the MLE minimises $KL(\hat{p} \parallel q_\theta)$. For 7 heads in 10 the minimum sits at $\theta = 0.7$, where the loss equals $H(\hat{p}) = 0.881$ bits. A one-hot label has $H = 0$ and loss $-\log q_\theta(y)$: $-\ln 0.7 = 0.357$ nats for output (0.2, 0.7, 0.1) and true class 2. Binary labels give $-\log q_\theta(y) = -\log(1 - \hat{y})$, the logistic regression loss (the MLE for linear and logistic regression sheet). The slides open this link on their last frame and stop.

11 Mutual information: bits one variable gives about another



$$I(X; Y) = H(X) - H(X \mid Y) = H(X) + H(Y) - H(X, Y) = KL(p(x, y) \parallel p(x)p(y)) \geq 0$$

$H(X \mid Y) = 0.35 \cdot 0.592 + 0.65 \cdot 0.891 = 0.786$: knowing about rain saves 0.214 of the sky's 1 bit. I is symmetric and 0 only for independent variables. A decision tree's information gain is I between label and attribute (the decision trees sheet). The slides stop before I .

12 Pause and think

Can a code average 1.5 bits on the halving source? No: $L \geq H = 1.75$.

Why does q 's Huffman code (2.3) cost less than $H(p, q) = 2.418$? It rounds the lengths $-\log_2 q_i$; $H(p, q)$ is not a floor.

A fair die? $\log_2 6 = 2.585$ bits, so 3-bit words waste 0.415.

$$H(p) = \mathbb{E}_p[-\log p] \leq L \cdot H(p, q) = H(p) + KL(p \parallel q), KL \geq 0 \cdot \text{cross-entropy loss} = \text{MLE} \cdot I(X; Y) = H(X) - H(X|Y)$$

Source: lecture slides "Information theory for machine learning" and the information theory notebook; the Gaussian pair is from the KL divergence notebook, nipunbatra.github.io/pml-teaching. Numbers: NumPy and exact fractions. The coding bound, the reverse Gaussian KL, the likelihood link and mutual information go beyond the slides. Corrections: the slides' Huffman code for q (A 10, B 0, C 110, D 111) should have lengths A 3, B 1, C 3, D 2, and its stated cost on p , 2.2 bits, is 2.0 by its own sum (2.3 with the right code); Huffman's length equals the entropy only for powers of 1/2 (1.9 against 1.8464 on the same slides); the KL 0.57148 rounds to 0.5715, not 0.5714.